
Crossed-Null Chronometry

Priority audit and dynamical viability test

Full field equations, stability, $3 + 1$ D realization and gravity coupling
with a dimensionless prediction

Audit conclusion. The original exact-null construction contains a correct and suggestive kinematic identity, but its conformal metric is algebraically the standard mimetic map and its first proposed exact-null action is transversely strongly coupled in $3 + 1$ dimensions. A nearby soft-null phase theory can be made ghost-free, gradient-stable and subluminal. A healthy heavy mediator generates the required sign and yields falsifiable speed and dispersion relations. A complete microscopic UV theory and a genuinely emergent gravitational metric are not yet obtained.

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Abstract

This report audits a proposal in which two compact null phases are used to reconstruct a timelike clock direction, a spacelike ruler direction and a conformal scale. The first result is negative but clarifying. Writing $\phi_{\pm} = T \pm R$, the exact null constraints imply $\nabla T \cdot \nabla R = 0$ and $(\nabla R)^2 = -(\nabla T)^2$. Consequently, the proposed metric

$$h_{ab} = C g_{ab}, \quad C = -\frac{1}{2} \nabla \phi_+ \cdot \nabla \phi_-,$$

reduces exactly to $h_{ab} = -(\nabla T)^2 g_{ab}$, the standard mimetic conformal map, supplemented by an equal-norm orthogonal ruler scalar. The kinematic map therefore does not support a priority claim. A systematic search also finds direct precedents for null-pair boost freedom, cross-flowing null-dust internal time, material clock/ruler fields, conformal gauge fixing by clock protocols, two-superfluid master functions, Bateman scalar representations of null electromagnetism and interaction-generated scales.

We then test the dynamics. The original Lagrange-multiplier theory enforcing two exactly null phase gradients has a rank-deficient quadratic principal part: transverse phase variations have no quadratic restoring term and enter only through nonlinear constraints. It is therefore not a conventional strongly hyperbolic 3+1-dimensional effective field theory about the proposed crossed-null background. We replace it by a soft-null phase EFT,

$$\mathcal{L} = -\frac{F}{2}(Y_+ + Y_-) + \frac{\kappa}{2}(C - m^2)^2,$$

where $Y_{\pm} = (\nabla \phi_{\pm})^2$. Its quadratic Hamiltonian is positive and both modes are real and subluminal when $F > 0$ and $0 < \epsilon = \kappa m^2 / (2F) < 1$. A local massive scalar mediator generates the positive operator coefficient and has a positive quadratic Hamiltonian precisely in the same domain. The exact spectrum contains one luminal mode, one anisotropic light mode and one heavy mode. The minimal theory predicts

$$c_{\parallel}^2 = 2c_{\perp}^2 - 1$$

and a second dimensionless relation linking the k^4 dispersion coefficient to the heavy gap. A hidden asymptotically free gauge sector can generate every dimensionful coefficient by dimensional transmutation, providing a credible 3 + 1-dimensional scale-generation module, but not yet a complete renormalizable UV completion of the entire crossed-phase sector. Finally, direct substitution of h_{ab} into the Einstein–Hilbert action is a singular mimetic construction; a regular Weyl-compensator lift gives a consistent gravity coupling but leaves conformal geometry fundamental. The viable conclusion is therefore narrower: causal/conformal structure may be primitive while local chronometric calibration is a collective field observable.

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1 Executive verdict

The requested programme was

priority search $\longrightarrow$ full equations $\longrightarrow$ ghost/gradients stability
 $\longrightarrow$ 3 + 1D realization $\longrightarrow$ gravity coupling $\longrightarrow$ prediction.

The outcome is summarized in Table 1.

Table 1: Research scorecard after the v0.2 audit.

Test	Status	Result
Systematic priority audit	PASS	The exact conformal map is mimetic, and the repaired two-phase EFT is a special two-superfluid master function. No exact match was located for the restricted crossed-null background, minimal mediator completion and the two derived sum rules taken together.
Original exact-null action	FAIL	The transverse phase sector is absent from the quadratic principal part. The homogeneous potential Hessian used in v0.1 did not test field-theory hyperbolicity.
Repaired soft-null EFT	PASS	Positive quadratic Hamiltonian, real frequencies and subluminal propagation for $0 < \epsilon < 1$.
Local mediator completion	PASS	A healthy massive scalar generates the positive lock coefficient. The full three-field Hamiltonian is positive for $\lambda^2 < FM^2$.
Microscopic 3+1D origin	PARTIAL	Dimensional transmutation and massive resonances are available in ordinary 3 + 1D QFT. A complete renormalizable model producing this exact compact two-phase sector has not yet been built.
Non-circular gravity coupling	PARTIAL	A regular Weyl-compensator/Einstein-frame coupling is consistent. A metric emerging solely from the phases is not established; the direct composite route is mimetic and singular.
Dimensionless prediction	PASS	The minimal model gives a tree-level angular sound-speed sum rule and a heavy-gap/dispersion relation. These are model-specific and receive corrections from additional operators.

The strongest defensible statement is now:

Causal/conformal order may be primitive, while chronometric calibration can be state-generated.

The report does *not* establish that spacetime, gravity or causal order itself emerges from two photons.

2 Systematic literature and priority audit

2.1 Search method

The search was designed as a priority falsification exercise rather than as a hunt for friendly citations. Targeted primary-literature queries were run across arXiv-indexed and INSPIRE-indexed material in twelve clusters:

1. double-null and $2 + 2$ formulations of general relativity;
2. null tetrads and reciprocal boost rescalings;
3. mimetic and multi-field mimetic gravity;
4. cross-flowing null dust and internal time;
5. dust clocks, scalar clock/ruler fields and relational observables;
6. emission coordinates and operational null coordinates;
7. relativistic superfluids, entrainment and two-stream instability;
8. self-gravitating media built from scalar fields;
9. EPS/Weyl reconstruction, clock protocols and second-clock effects;
10. causal-order and order-plus-number reconstruction;
11. dimensional transmutation and confining mass gaps;
12. positivity bounds and mediator-generated derivative operators;
13. Bateman scalar potentials for null electromagnetism and principal observers of non-null electromagnetic fields.

Exact-combination searches included “crossed null phases”, “two null phases conformal metric”, “null phase clock”, “eikonal scalar fields conformal factor”, and “double-null clock ruler fields”. The accompanying CSV records the closest primary papers, overlap, residual distinction and priority risk. The absence of an exact hit is not proof of priority; non-arXiv journals, older terminology and unpublished work remain possible.

2.2 What is clearly established

The following ingredients are prior art.

Null-pair algebra. A cross-normalized pair of null directions can be combined into orthogonal timelike and spacelike directions, and reciprocal rescaling of the null pair is boost freedom. This is standard null-frame geometry and appears throughout Newman–Penrose and $2 + 2$ formalisms; recent work makes the null/timelike correspondence explicit [1, 2].

Double-null scale freedom. Double-null formulations contain a cross-normalization or null lapse, and a restricted Weyl symmetry can be made manifest after choosing the parametrization of null generators [3]. Thus the general association among null normalization, boost freedom and conformal scale is not new.

Relational time from two null streams. Cross-flowing null dust is equivalent to an anisotropic reference fluid possessing a preferred internal time; that time can linearize the gravitational Hamiltonian constraint [9, 10]. The slogan “one null stream has no rest time, two nonparallel streams can” therefore has a direct precedent.

Material clocks and rulers. Dust proper time and comoving labels can be used as canonical spacetime references [11]. Scalar clock/ruler fields and composite metrics have also been used in emergent-gravity and relational-observable programmes [13, 14]. Four broadcast clock readings define null emission coordinates in relativistic positioning systems [12].

Scalar-media dynamics. Derivatively coupled scalars already provide a general EFT language for fluids, superfluids, solids and supersolids [15]. More decisively, the general zero-temperature two-superfluid constitutive law is an arbitrary generalized pressure

$$\Psi = \Psi(p_1^2, p_2^2, p_{12}^2), \quad p_{12}^2 = p_1 \cdot p_2,$$

and microscopic two-complex-scalar models already generate entrainment interactions and angle-dependent sound spectra [16, 17]. The repaired crossed-phase Lagrangian is therefore a polynomial special case of an established two-superfluid master-function class, evaluated on a special crossed-null state. Its theory class, two-phase operator architecture and generic stability problem are not novel.

Literal electromagnetism. Bateman’s construction uses two complex scalar potentials satisfying a first-order compatibility equation to generate null Maxwell fields [18]. For non-null electromagnetic fields, principal observers with vanishing Poynting vector exist and lie in the timelike plane selected by the principal null directions [19]. These results preclude a broad priority claim based merely on “two scalar/null structures” or on a non-null electromagnetic superposition acquiring an energy rest frame. Bateman potentials are nevertheless not two independent compact phases whose gradients are separately null, so they do not reproduce the model tested below.

Clocks and conformal gauge. In EPS/Weyl geometry, a congruence of standard clocks can select a representative of a conformal class, and clock-history dependence is constrained by the second-clock-effect problem [20, 21]. The move from conformal causal structure to a clock-calibrated metric is established at the operational level.

State-selected and scale-generated time. The thermal-time hypothesis makes physical time flow state-dependent [22]. Dimensional transmutation produces physical mass scales from classically scale-free interactions [27, 28]. Neither idea is new separately.

2.3 The decisive priority result: exact reduction to mimetic gravity

Let

$$\phi_+ = T + R, \quad \phi_- = T - R,$$

and abbreviate $T_a = \nabla_a T$, $R_a = \nabla_a R$. The two exact null constraints are

$$0 = (T + R)^2 = T^2 + R^2 + 2T \cdot R, \quad (1)$$

$$0 = (T - R)^2 = T^2 + R^2 - 2T \cdot R. \quad (2)$$

Adding and subtracting gives

$$\boxed{T \cdot R = 0, \quad R^2 = -T^2.} \quad (3)$$

The crossed invariant is

$$C = -\frac{1}{2} \nabla \phi_+ \cdot \nabla \phi_- \quad (4)$$

$$= -\frac{1}{2} (T + R) \cdot (T - R) \quad (5)$$

$$= -\frac{1}{2} (T^2 - R^2) \quad (6)$$

$$= -T^2 = R^2. \quad (7)$$

Consequently,

$$\boxed{h_{ab} = C g_{ab} = -\left(g^{cd} T_c T_d\right) g_{ab}.} \quad (8)$$

Equation (8) is precisely the original mimetic conformal transformation, with T playing the mimetic scalar [4]. The second scalar R supplies an orthogonal equal-norm ruler direction but does not make the conformal map new.

Proposition 2.1 (Updated priority statement). *The exact crossed-null “conformal completion” is an algebraic corollary of the mimetic map plus the two null constraints. Its theorem-level content must not be claimed as new.*

This also raises a dynamical warning. Multi-field mimetic models possess additional scalar modes, and their health depends sensitively on the field-space signature; opposite-sign clock/ruler constructions can carry ghosts [5, 6]. Extended mimetic models commonly develop gradient instabilities, ghost modes or strong coupling unless additional structure is supplied [7, 8].

2.4 Narrow candidate novelty after the search

The two-superfluid literature already contains the general invariant dependence on p_1^2 , p_2^2 and $p_1 \cdot p_2$. Consequently, novelty cannot attach to the existence of two compact phases, entrainment, relative-flow anisotropy, angular sound modes or a derivative interaction generated by heavy exchange. Within the targeted search, no exact primary-source match was located for the following narrower conjunction:

1. a two-phase master function evaluated specifically on a crossed-null background;
2. the explicit diagnosis that exact null constraints leave a transversely degenerate quadratic principal symbol on that background;
3. the minimal soft-null polynomial and its one-scalar local mediator completion;
4. the two restricted tree-level consistency relations derived in Sections 6 and 10.

This conjunction is a **CANDIDATE** contribution, not a settled or legal priority claim. The search can falsify overbroad novelty claims, but absence of an indexed exact hit cannot prove originality.

Before submission, backward and forward citation chasing from the mimetic, multi-superfluid, Bateman and self-gravitating-media papers, non-arXiv database searches and specialist review remain necessary.

3 Kinematic core that remains useful

Although the exact conformal map is not new, the null decomposition remains a useful organizing language.

Let k_a and ℓ_a be future-directed, nonparallel null covectors with

$$k^2 = \ell^2 = 0, \quad k \cdot \ell < 0.$$

For a chosen cross-normalization $-k \cdot \ell = 2$, define

$$u_a = \frac{k_a + \ell_a}{2}, \quad e_a = \frac{k_a - \ell_a}{2}. \quad (9)$$

Then

$$u^2 = -1, \quad e^2 = +1, \quad u \cdot e = 0.$$

Under reciprocal rescaling

$$k_a \mapsto e^\eta k_a, \quad \ell_a \mapsto e^{-\eta} \ell_a,$$

one obtains

$$u_a \mapsto \cosh \eta u_a + \sinh \eta e_a, \quad (10)$$

$$e_a \mapsto \sinh \eta u_a + \cosh \eta e_a. \quad (11)$$

Thus η is rapidity.

More generally, under independent positive rescalings

$$k \mapsto ak, \quad \ell \mapsto b\ell,$$

define

$$\alpha = \frac{1}{2} \ln(ab), \quad \eta = \frac{1}{2} \ln(a/b).$$

The ratio controls boost; the product controls common scale. This decomposition is conceptually useful even though it is not a new mathematical result:

relative null normalization = rapidity,	geometric mean normalization = clock scale.
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(12)

For physical momenta p and q rather than unscaled null directions,

$$P^a = p^a + q^a, \quad M^2 = -P^2 = -2p \cdot q.$$

Whenever p and q are nonparallel, P is timelike and

$$u^a = \frac{P^a}{M}$$

is the pair's centre-of-momentum time direction. The physical energies fix the relative normalization that abstract null rays lack.

4 Original exact-null field theory: full equations and failure

4.1 Action

A concrete completion of the v0.1 proposal is

$$S_0 = \int d^4x \sqrt{-g} \left[\frac{M_{\text{P}}^2}{2} R + A_+ Y_+ + A_- Y_- - \frac{\kappa}{2} (C - \sigma^2)^2 - \frac{1}{2} (\nabla \sigma)^2 - V(\sigma) \right], \quad (13)$$

where

$$Y_+ = g^{ab} \nabla_a \phi_+ \nabla_b \phi_+, \quad Y_- = g^{ab} \nabla_a \phi_- \nabla_b \phi_-, \quad C = -\frac{1}{2} g^{ab} \nabla_a \phi_+ \nabla_b \phi_-,$$

and $A_{\pm}$ are Lagrange multipliers.

Set

$$p_a = \nabla_a \phi_+, \quad q_a = \nabla_a \phi_-, \quad D = C - \sigma^2.$$

4.2 Full Euler–Lagrange equations

Variation with respect to the multipliers gives the exact null constraints

$$\boxed{Y_+ = 0, \quad Y_- = 0.} \quad (14)$$

The phase equations are

$$\nabla_a \left(2A_+ p^a + \frac{\kappa}{2} D q^a \right) = 0, \quad (15)$$

$$\nabla_a \left(2A_- q^a + \frac{\kappa}{2} D p^a \right) = 0. \quad (16)$$

The scale-field equation is

$$\boxed{\square \sigma + 2\kappa \sigma D - V'(\sigma) = 0.} \quad (17)$$

The Einstein equation is

$$M_{\text{P}}^2 G_{ab} = T_{ab}^{(0)},$$

with

$$T_{ab}^{(0)} = -2A_+ p_a p_b - 2A_- q_a q_b - \frac{\kappa D}{2} (p_a q_b + p_b q_a) + \nabla_a \sigma \nabla_b \sigma + g_{ab} \mathcal{L}_0. \quad (18)$$

Equations (14)–(18) are the complete classical field equations for the minimal exact-null model.

4.3 Crossed background

In local Minkowski coordinates, consider

$$\bar{\phi}_+ = m(t - x), \quad \bar{\phi}_- = m(t + x), \quad \bar{\sigma} = m, \quad (19)$$

with $\bar{A}_{\pm} = 0$, $V(m) = V'(m) = 0$. Then

$$\bar{Y}_+ = \bar{Y}_- = 0, \quad \bar{C} = m^2, \quad \bar{D} = 0.$$

This solves the local matter equations and has vanishing stress in the tuned background.

4.4 Linearized constraints

Write

$$\phi_{\pm} = \bar{\phi}_{\pm} + \pi_{\pm}.$$

The linearized null constraints are

$$(\partial_t + \partial_x)\pi_+ = 0, \tag{20}$$

$$(\partial_t - \partial_x)\pi_- = 0. \tag{21}$$

Introduce sum and difference perturbations

$$s = \frac{\pi_+ + \pi_-}{\sqrt{2}}, \quad r = \frac{\pi_+ - \pi_-}{\sqrt{2}}.$$

The linear perturbation of C is

$$\delta C = \frac{m}{\sqrt{2}}(\dot{s} - \partial_x r). \tag{22}$$

Crucially, neither the constraints nor δC contains ∂_y or ∂_z at linear order.

4.5 Transverse strong coupling

Take a transverse perturbation with $k_x = 0$. Equations (20)–(21) force $\dot{\pi}_{\pm} = 0$ at linear order, while the quadratic lock contains no transverse gradient. The first transverse dependence enters through nonlinear terms in the exact null constraints, schematically

$$Y_{\pm} = 2\bar{p}_{\pm} \cdot \nabla \pi_{\pm} + (\nabla \pi_{\pm})^2,$$

so $(\partial_y \pi_{\pm})^2$ and $(\partial_z \pi_{\pm})^2$ appear one perturbative order later than the longitudinal principal terms.

Failure mode 4.1 (Rank-deficient principal part). *The linearized 3 + 1-dimensional system does not possess a nondegenerate quadratic principal symbol for generic transverse phase perturbations. Those modes have no quadratic restoring energy and become controlled by nonlinear terms. In effective-field-theory terms this is a strong-coupling warning: nonlinear terms control directions that have no quadratic restoring operator. A hidden constraint structure could alter that diagnosis, but none has been exhibited for the proposed action.*

A positive Hessian of the homogeneous potential in (C, σ) does not test this issue. It checks only algebraic fluctuations of homogeneous order parameters, not kinetic rank, gradient signs or hyperbolicity.

Verdict on the v0.1 exact-null dynamics: FAIL. The kinematic constraints are elegant, but exact nullity removes the transverse quadratic dynamics required for a controlled 3 + 1D EFT.

5 Repaired soft-null phase EFT

The minimal repair is to make the phases ordinary propagating fields and treat the crossed-null configuration as a background state rather than as an exact pointwise constraint.

5.1 Action and field equations

Consider

$$S_{\text{EFT}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{P}}^2}{2} R - \frac{F}{2} (Y_+ + Y_-) + \frac{\kappa}{2} (C - m^2)^2 \right], \quad (23)$$

with

$$F > 0, \quad \kappa > 0.$$

Here the phases are dimensionless, so F and m^2 have mass dimension two and κ is dimensionless. Let $D = C - m^2$. The phase equations are

$$\nabla_a \left(F p^a + \frac{\kappa}{2} D q^a \right) = 0, \quad (24)$$

$$\nabla_a \left(F q^a + \frac{\kappa}{2} D p^a \right) = 0. \quad (25)$$

The stress tensor is

$$T_{ab}^{\text{EFT}} = F(p_a p_b + q_a q_b) + \frac{\kappa D}{2} (p_a q_b + p_b q_a) + g_{ab} \mathcal{L}_{\text{EFT}}. \quad (26)$$

Together with

$$M_{\text{P}}^2 G_{ab} = T_{ab}^{\text{EFT}},$$

these are the complete classical field equations.

The crossed configuration (19) with fixed m solves the phase equations in flat space. Unlike the exact-null model, however, it carries stress

$$\bar{T}_{ab} = F(\bar{p}_a \bar{p}_b + \bar{q}_a \bar{q}_b).$$

Thus flat spacetime is a decoupling-limit or local laboratory approximation, not a self-consistent global gravitating vacuum. One crossed pair is anisotropic. Three equal orthogonal crossed pairs produce isotropic radiation-like stress with $w = 1/3$, but still gravitate.

5.2 Quadratic action

Perturb around (19) and use s, r as above. Define

$$a = \frac{\kappa m^2}{2}. \quad (27)$$

Since δC is given by (22), the quadratic Lagrangian is

$$\begin{aligned} \mathcal{L}^{(2)} = & \frac{1}{2} (F + a) \dot{s}^2 + \frac{1}{2} F \dot{r}^2 - a \dot{s} \partial_x r \\ & - \frac{F}{2} [(\nabla s)^2 + (\nabla r)^2] + \frac{a}{2} (\partial_x r)^2. \end{aligned} \quad (28)$$

The canonical momenta are

$$P_s = (F + a) \dot{s} - a \partial_x r, \quad P_r = F \dot{r}. \quad (29)$$

The Hamiltonian density is

$$\mathcal{H}^{(2)} = \frac{(P_s + a\partial_x r)^2}{2(F + a)} + \frac{P_r^2}{2F} + \frac{F}{2} [(\nabla s)^2 + (\nabla r)^2] - \frac{a}{2}(\partial_x r)^2. \quad (30)$$

Theorem 5.1 (Quadratic Hamiltonian positivity). *The Hamiltonian (30) is positive definite for nonzero perturbations if*

$$\boxed{F > 0, \quad 0 < a < F.} \quad (31)$$

Proof. The P_r and transverse-gradient terms are positive for $F > 0$. The nontrivial block in $(P_s, \partial_x r)$ has, apart from an overall positive factor, determinant

$$\det H_{(P_s, r_x)} = \frac{F - a}{F + a}.$$

It is positive precisely when $a < F$. The leading principal minor is positive when $F + a > 0$, automatically satisfied in (31). $\square$

5.3 Dispersion relation

The linear equations are

$$(F + a)\ddot{s} - a\partial_x \dot{r} - F\nabla^2 s = 0, \quad (32)$$

$$F\ddot{r} - a\partial_x \dot{s} - (F - a)\partial_x^2 r - F\nabla_\perp^2 r = 0. \quad (33)$$

For $e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}$, the determinant factorizes exactly:

$$F(\omega^2 - k^2) \left[(F + a)\omega^2 - Fk^2 + ak_x^2 \right] = 0. \quad (34)$$

The two branches are

$$\omega_1^2 = k^2, \quad (35)$$

$$\omega_2^2 = \frac{Fk^2 - ak_x^2}{F + a}. \quad (36)$$

Define

$$\epsilon = \frac{a}{F} = \frac{\kappa m^2}{2F}, \quad 0 < \epsilon < 1, \quad (37)$$

and let θ be the angle between $\mathbf{k}$ and the crossed-null axis. Then

$$\boxed{c_2^2(\theta) = \frac{1 - \epsilon \cos^2 \theta}{1 + \epsilon}.} \quad (38)$$

Every frequency is real and every phase speed is at most one in the healthy domain. There is no ghost, no gradient instability and no superluminal low-energy branch.

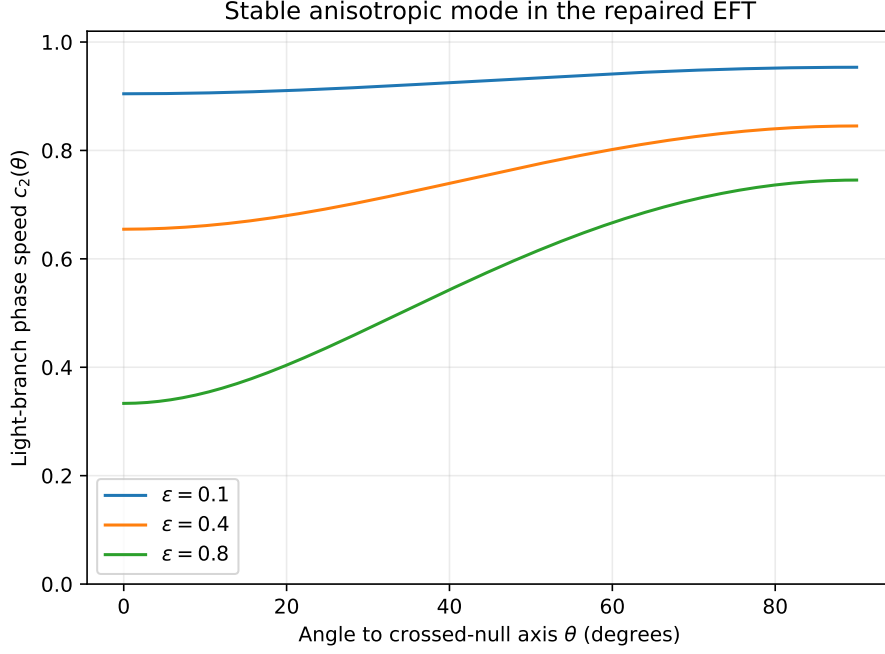


Figure 1: Low-energy phase speed of the anisotropic branch. Increasing ϵ slows the mode and increases its directional anisotropy, while the branch remains real and subluminal for $0 < \epsilon < 1$.

5.4 Why the sign changed from v0.1

With a conventional canonical phase kinetic term, the v0.1 sign $-\kappa D^2/2$ produces a low-energy branch whose speed is generically superluminal in the algebraically stable domain. The positive sign in (23) is the sign generated by healthy massive exchange and is compatible with the standard positivity logic for derivative operators [25, 26]. Gravity complicates rigorous forward-limit positivity bounds, so this is an EFT diagnostic rather than a theorem about quantum gravity.

Verdict on the repaired phase EFT: PASS at quadratic order. It is a controlled local phase medium for $0 < \epsilon < 1$. Nonlinear stability, loops and curved backgrounds remain to be tested.

6 Local 3 + 1D mediator completion

6.1 Action and exact equations

Introduce a real scalar mediator H :

$$S_{\text{med}} = \int d^4x \sqrt{-g} \left[\frac{M_P^2}{2} R - \frac{F}{2} (Y_+ + Y_-) - \frac{1}{2} (\nabla H)^2 - \frac{1}{2} M^2 H^2 + g_H H (C - m^2) \right]. \quad (39)$$

The phase equations are

$$\nabla_a \left(F p^a + \frac{g_H H}{2} q^a \right) = 0, \quad (40)$$

$$\nabla_a \left(F q^a + \frac{g_H H}{2} p^a \right) = 0. \quad (41)$$

The mediator equation is

$$(M^2 - \square)H = g_H(C - m^2). \quad (42)$$

The stress tensor is

$$T_{ab}^{\text{med}} = F(p_a p_b + q_a q_b) + \nabla_a H \nabla_b H + \frac{g_H H}{2}(p_a q_b + p_b q_a) + g_{ab} \mathcal{L}_{\text{med}}. \quad (43)$$

Together with $M_{\text{p}}^2 G_{ab} = T_{ab}^{\text{med}}$, these are the full field equations.

At energies below M , integrating out H gives

$$\Delta \mathcal{L}_{\text{eff}} = \frac{g_H^2}{2}(C - m^2) \frac{1}{M^2 - \square}(C - m^2) \quad (44)$$

$$= \frac{g_H^2}{2M^2}(C - m^2)^2 + \frac{g_H^2}{2M^4}(C - m^2)\square(C - m^2) + \dots \quad (45)$$

Thus

$$\kappa = \frac{g_H^2}{M^2} > 0. \quad (46)$$

The stable sign is not inserted by hand; it follows from healthy exchange.

6.2 Quadratic Hamiltonian

Around the crossed background, define

$$\lambda = \frac{g_H m}{\sqrt{2}}. \quad (47)$$

The quadratic Lagrangian is

$$\mathcal{L}_2 = \frac{F}{2} \left(\dot{s}^2 + \dot{r}^2 - (\nabla s)^2 - (\nabla r)^2 \right) + \frac{1}{2} \left(\dot{H}^2 - (\nabla H)^2 - M^2 H^2 \right) + \lambda H(\dot{s} - \partial_x r). \quad (48)$$

The momenta are

$$P_s = F\dot{s} + \lambda H, \quad P_r = F\dot{r}, \quad P_H = \dot{H}.$$

The Hamiltonian can be arranged as

$$\begin{aligned} \mathcal{H}_2 = & \frac{(P_s - \lambda H)^2}{2F} + \frac{P_r^2}{2F} + \frac{P_H^2}{2} + \frac{F}{2}(\nabla s)^2 + \frac{F}{2}(\nabla_{\perp} r)^2 + \frac{1}{2}(\nabla H)^2 \\ & + \frac{F}{2} \left(\partial_x r + \frac{\lambda}{F} H \right)^2 + \frac{1}{2} \left(M^2 - \frac{\lambda^2}{F} \right) H^2. \end{aligned} \quad (49)$$

Therefore

$$F > 0, \quad M^2 > 0, \quad \lambda^2 < FM^2 \quad (50)$$

is sufficient and necessary for positivity of the quadratic Hamiltonian. Since

$$\epsilon = \frac{\lambda^2}{FM^2},$$

this is exactly the EFT condition $0 < \epsilon < 1$.

6.3 Exact spectrum

The Fourier determinant factorizes as

$$(\omega^2 - k^2) \left[F(k^2 - \omega^2)(M^2 + k^2 - \omega^2) - \lambda^2(\omega^2 + k_x^2) \right] = 0. \quad (51)$$

Besides the exact luminal branch, define

$$\mu^2 = \frac{\lambda^2}{F}.$$

The two mixed branches are

$$\omega_{\pm}^2 = \frac{M^2 + \mu^2 + 2k^2 \pm \sqrt{(M^2 + \mu^2)^2 + 4\mu^2(k^2 + k_x^2)}}{2}. \quad (52)$$

At $k = 0$,

$$\omega_-^2 = 0, \quad \omega_+^2 = \Omega_H^2 = M^2 + \mu^2 = M^2(1 + \epsilon).$$

The low-energy light branch is

$$\omega_-^2 = \frac{k^2 - \epsilon k_x^2}{1 + \epsilon} + \frac{\epsilon^2}{M^2(1 + \epsilon)^3} (k^2 + k_x^2)^2 + O(k^6/M^4). \quad (53)$$

The leading term reproduces (36); the higher terms resolve the EFT near the mediator scale.

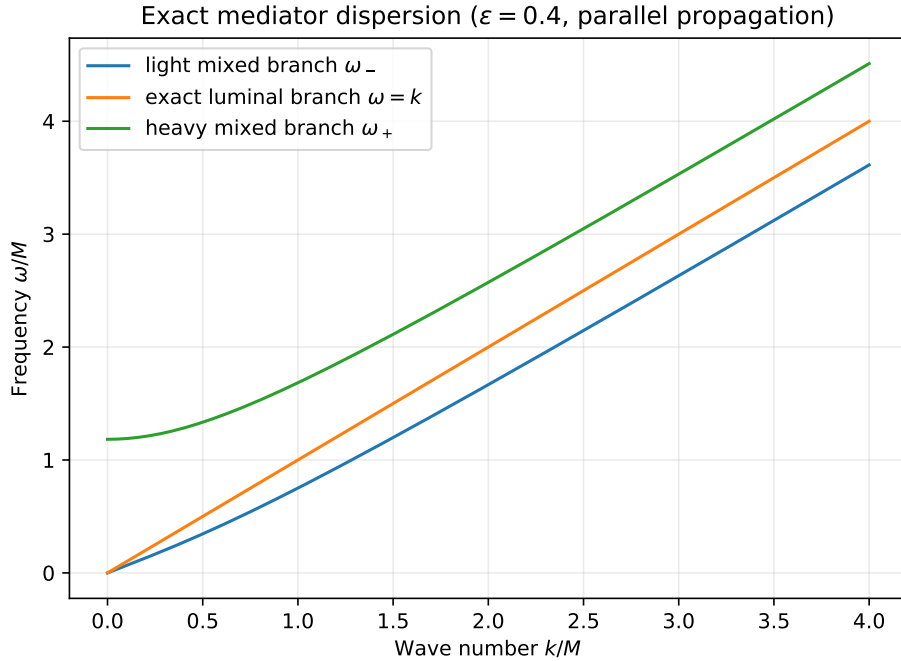


Figure 2: Exact branches of the local mediator model for parallel propagation and $\epsilon = 0.4$. The full theory contains an exact luminal mode, an anisotropic light mode and a heavy mode.

7 Microscopic 3 + 1D realization: what passes and what does not

7.1 What “microscopic realization” must mean

A genuine realization must supply, rather than merely name:

1. two compact low-energy phase variables;
2. a state containing oppositely directed or nonparallel phase gradients;
3. a positive cross-derivative interaction;
4. a physical scale generated without inserting a classical mass by hand;
5. a cutoff above the clock/phase frequencies.

The mediator model supplies items 2–3 as a local 3 + 1D field theory, but its coefficients F, m, M, g_H are still parameters. Dimensional transmutation can supply item 4.

7.2 Hidden asymptotically free sector

Let a hidden non-Abelian gauge coupling $g_*(\mu)$ obey

$$\mu \frac{dg_*}{d\mu} = \beta(g_*) = -b_0 g_*^3 + O(g_*^5), \quad b_0 > 0.$$

The RG-invariant scale is

$$\Lambda_* = \mu \exp \left[- \int^{g_*(\mu)} \frac{dg}{\beta(g)} \right] \simeq \mu \exp \left[- \frac{1}{2b_0 g_*^2(\mu)} \right]. \quad (54)$$

Classically massless four-dimensional gauge theories can therefore generate massive spectra and condensates through quantum running, as in QCD [28]. Hidden Yang–Mills sectors with glueball spectra are standard concrete examples [29].

Parameterize the low-energy coefficients by dimensionless constants:

$$m = \zeta \Lambda_*, \quad F = f^2 \Lambda_*^2, \quad M = y \Lambda_*, \quad g_H = \gamma \Lambda_*. \quad (55)$$

Then

$$\epsilon = \frac{g_H^2 m^2}{2F M^2} = \frac{\gamma^2 \zeta^2}{2f^2 y^2}, \quad (56)$$

and the stability condition is a pure dimensionless inequality.

7.3 Origin of compact phases

Two compact phases can arise as Goldstone modes of two broken global $U(1)$ symmetries or as commuting phases in a non-Abelian condensate. Their ordinary two-derivative kinetic term provides F . Heavy radial or composite resonances generate higher-derivative phase interactions after being integrated out. This is the standard architecture of superfluid and chiral EFTs. In fact, the resulting phase action is a restricted choice of the established two-superfluid generalized pressure $\Psi(p_1^2, p_2^2, p_{12}^2)$ and also lies in the broader self-gravitating-media class [15, 16]. The 3 + 1D realization task is therefore not to invent a new species of EFT; it is to produce a technically

natural microscopic model whose vacuum or state realizes the crossed-null background and whose matching enforces the minimal coefficient relations used here.

What is not yet supplied is a fully renormalizable UV Lagrangian whose matching produces exactly the single operator in (23), suppresses every competing operator enough to preserve the sum rules, and dynamically prepares the crossed-null state. The local term $H \nabla \phi_+ \cdot \nabla \phi_-$ is a legitimate low-energy interaction but is not by itself a proof of an all-scale renormalizable completion.

Microscopic verdict: PARTIAL. Four-dimensional dimensional transmutation, compact phases and massive mediator resonances are individually real. Their exact combination has a controlled local EFT completion, but not yet a complete microscopic UV model.

8 Gravity coupling and the circularity test

8.1 Route A: substitute the phase metric into Einstein–Hilbert

One might propose

$$S_{\text{grav}}[g, \phi_{\pm}] = \frac{M_{\text{P}}^2}{2} \int \sqrt{-h} R[h], \quad h_{ab} = C g_{ab}.$$

Under the exact null constraints, however, h_{ab} is the mimetic map (8). The transformation is noninvertible: a Weyl rescaling of the auxiliary metric leaves h_{ab} unchanged. Substitution into curvature therefore creates the same extra conformal dynamics and constraint structure as mimetic gravity [4]. The known instability and strong-coupling literature applies directly enough that this cannot be advertised as a clean emergent-gravity mechanism [5, 7].

Direct composite-metric route: FAIL as the preferred formulation. It is singular, noninvertible and not a new escape from mimetic gravity.

8.2 Route B: regular Weyl-compensator lift

A consistent alternative starts with a conformal metric $[g]$ and a compensator χ of Weyl weight -1 :

$$g_{ab} \mapsto \Omega^2 g_{ab}, \quad \chi \mapsto \Omega^{-1} \chi.$$

Define the invariant physical metric

$$\hat{g}_{ab} = \frac{\chi^2}{6M_{\text{P}}^2} g_{ab}. \quad (57)$$

The Einstein–Hilbert action is equivalently

$$\frac{M_{\text{P}}^2}{2} \int \sqrt{-\hat{g}} R[\hat{g}] = \int \sqrt{-g} \left[\frac{\chi^2}{12} R[g] + \frac{1}{2} (\nabla \chi)^2 \right] \quad (58)$$

up to a boundary term. The apparent wrong-sign compensator mode is pure Weyl gauge in this redundant parametrization.

A classically Weyl-invariant phase sector is

$$S_W = \int \sqrt{-g} \left[\frac{\chi^2}{12} R + \frac{1}{2} (\nabla \chi)^2 - \frac{f^2 \chi^2}{2} (Y_+ + Y_-) + \frac{\kappa}{2} (C - \zeta \chi^2)^2 \right]. \quad (59)$$

The phase equations are

$$\nabla_a \left(f^2 \chi^2 p^a + \frac{\kappa}{2} D_W q^a \right) = 0, \quad (60)$$

$$\nabla_a \left(f^2 \chi^2 q^a + \frac{\kappa}{2} D_W p^a \right) = 0, \quad (61)$$

where $D_W = C - \zeta \chi^2$. The compensator equation is

$$-\square \chi + \frac{R}{6} \chi - f^2 \chi (Y_+ + Y_-) - 2\kappa \zeta \chi D_W = 0. \quad (62)$$

The metric equation is equivalently the Einstein equation written for the invariant metric (57); no singular change of variables is required.

On the locking surface $C = \zeta \chi^2$,

$$C g_{ab} = 6\zeta M_{\text{P}}^2 \hat{g}_{ab}. \quad (63)$$

Thus the phase cross-normalization can calibrate the same metric used by gravity without replacing that metric by a noninvertible composite.

This is consistent but conceptually narrower. The conformal geometry and a gravitational action remain primitive. The phases select or track the chronometric gauge; they do not derive the Einstein dynamics from null incidence alone. Quantum Weyl anomalies and the generation of a hierarchy between χ and the phase scale also require a microscopic treatment [27, 30].

8.3 Backreaction and global time

A local clock medium can be treated as a test system. A universal condensate cannot. Its stress tensor would alter the spacetime it is supposed to calibrate. One crossed pair selects a spatial axis; an isotropic triad behaves like radiation and is tightly constrained cosmologically. A successful global theory would need one of the following:

- a localized operational clock sector rather than a universal medium;
- a stealth configuration whose stress cancels without reintroducing ghosts;
- a cosmological solution whose energy density is observationally acceptable;
- a discrete/order-theoretic formulation where the phase data calibrate event density rather than fill spacetime as ordinary matter.

Furthermore, a local timelike eigenvector field need not be hypersurface-orthogonal. If it has vorticity, local clocks exist but do not define one global cosmic “now”. That is not a defect; general relativity itself generically lacks a preferred global time.

9 Clock universality

Suppose clock species A has a local transition gap $\Delta E_A(x)$. A single metric time exists only if all good clocks share one local scale field:

$$\boxed{\Delta E_A(x) = c_A \sigma(x)}, \quad (64)$$

where c_A is a species-dependent constant. Equivalently,

$$\boxed{\mathrm{d} \ln \left(\frac{\Delta E_A}{\Delta E_B} \right) = 0} \quad (65)$$

for every pair of ideal clocks.

If (65) fails, the construction does not yield one universal chronometric metric. It predicts differential clock drift, a second-clock effect or an equivalence-principle violation. This criterion turns the philosophical phrase “time emerges” into a stringent empirical condition.

For a clock transition,

$$\Delta\varphi_A = \frac{1}{\hbar} \int \Delta E_A \mathrm{d}\tau.$$

Under (64), phase counts of every clock species infer the same $\sigma \mathrm{d}\tau$ up to fixed conversion constants. Chronometric duration is then relational but universal.

10 New dimensionless predictions of the minimal repaired model

10.1 Angular speed sum rule

From (38), propagation parallel and perpendicular to the crossed-null axis gives

$$c_{\parallel}^2 = \frac{1 - \epsilon}{1 + \epsilon}, \quad c_{\perp}^2 = \frac{1}{1 + \epsilon}. \quad (66)$$

Eliminating ϵ yields

$$\boxed{c_{\parallel}^2 = 2c_{\perp}^2 - 1.} \quad (67)$$

This contains no dimensionful parameter. It is a tree-level consistency relation of the minimal one-operator model.

10.2 Heavy-gap/dispersion relation

Write the light branch as

$$\omega_{-}^2 = c_2^2(\theta)k^2 + \alpha(\theta)k^4 + O(k^6). \quad (68)$$

Using (53),

$$\alpha(\theta) = \frac{\epsilon^2}{M^2(1 + \epsilon)^3} (1 + \cos^2 \theta)^2.$$

Since the measurable heavy gap is

$$\Omega_H^2 = M^2(1 + \epsilon),$$

one obtains the dimensionless relation

$$\boxed{\Omega_H^2 \alpha(\theta) = (1 - c_{\perp}^2)^2 (1 + \cos^2 \theta)^2.} \quad (69)$$

In particular,

$$\Omega_H^2 \alpha_{\perp} = (1 - c_{\perp}^2)^2, \quad (70)$$

$$\Omega_H^2 \alpha_{\parallel} = 4(1 - c_{\perp}^2)^2. \quad (71)$$

Equations (67) and (69) are jointly stronger than merely observing anisotropic propagation.

10.3 Status of the prediction

The prediction is new only in a restricted sense: it is the output of the minimal repaired Lagrangian, and no exact match was found in the targeted search. It is not protected against every allowed higher-derivative or multi-field operator. Loops, additional resonances and curved backgrounds will perturb it. A publishable claim should therefore read:

The minimal crossed-phase mediator EFT predicts a pair of tree-level consistency relations; their violation measures the size of operators beyond the minimal model.

Potential test beds include analogue two-component condensates, engineered nonlinear optical media and any hidden-sector phenomenology in which the phase modes couple weakly to observable fields. No present experiment has been identified as already constraining the model.

11 What “time falls out” means after the audit

The construction supports a precise but limited emergence chain:

causal/conformal structure $\longrightarrow$ before/after and null directions, nonparallel phase gradients $\longrightarrow$ a local timelike energy frame, interaction-generated scale $\longrightarrow$ a clock calibration, relative phase cycles $\longrightarrow$ elapsed duration.	(72)
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What does not yet fall out is equally important:

- the existence of events or causal order;
- the full gravitational dynamics;
- a global time foliation;
- the thermodynamic arrow of time;
- the universality of every matter clock;
- a UV-complete microscopic crossed-phase theory.

The most natural interpretation is therefore not “time is made of photons”. It is:

duration is not primitive bookkeeping; it is a calibrated relation among physical processes.	(73)
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This is provocative, but it is also compatible with the sober operational content of relativity.

12 Next research programme

The following sequence would convert the present audit into a submission-quality theory paper.

1. **General operator basis.** Classify the complete shift-symmetric two-phase EFT through four derivatives on a crossed-null background. Determine which operators preserve or deform (67).
2. **Nonlinear well-posedness.** Compute the full characteristic matrix and establish symmetric or strong hyperbolicity in the healthy domain.
3. **Loop analysis.** Calculate one-loop renormalization of F , κ and the competing four-derivative operators. Identify technically natural symmetry restrictions.
4. **Microscopic matching.** Build a concrete hidden confining or linear-sigma model, integrate out its radial/resonance spectrum, and calculate the sign and size of the crossed operator.
5. **Curved backgrounds.** Derive perturbations on FLRW and stationary spacetimes, including backreaction and vorticity of the energy frame.
6. **Clock universality.** Couple at least two distinct clock species and test (65) explicitly.
7. **Phenomenology.** Choose either an analogue system or a dark-sector portal and translate (67)–(69) into measured frequencies.
8. **Priority review.** Perform citation-network searches and solicit review from experts in mimetic gravity, relativistic superfluids and EFT positivity before asserting novelty.

13 Conclusion

The audit improves the project by killing the weakest version of it.

The exact phase metric is not a new conformal-completion theorem; it is the mimetic map written in crossed-null variables. The exact-null Lagrange-multiplier action is not a healthy $3 + 1$ D EFT around the proposed background because its transverse fluctuations are strongly coupled. Those are substantive negative results.

The underlying intuition nevertheless survives in a sharper form. A soft-null two-phase medium admits a positive Hamiltonian and stable subluminal propagation. A healthy mediator generates the required sign and produces an exact local spectrum. Four-dimensional dimensional transmutation can supply the scale. The minimal model makes dimensionless, falsifiable predictions.

The remaining frontier is no longer “can two null directions be called a clock?” That is established kinematics. The real question is:

Can a microscopic, universal and backreaction-consistent phase medium generate the same chronometric scale for all matter without importing a metric clock through its assumptions?

The present report gives a controlled EFT in which that question can now be asked mathematically.

A Symbolic checks

The accompanying Python/SymPy script verifies:

1. the reduction $C = -T^2 = R^2$ and the mimetic identity;
2. factorization of the repaired EFT determinant;

3. positivity of the Hamiltonian block for $0 < a < F$;
4. the exact mediator determinant and roots;
5. the k^4 expansion of the light branch;
6. both dimensionless predictions;
7. 20,000 random points in the healthy parameter domain with no negative ω^2 ;
8. 2,000 points outside $\epsilon < 1$, all showing the expected long-wavelength parallel tachyon.

The machine-readable result is supplied as `verification_results_v0_2.json`.

B Condensed priority matrix

Table 2: Closest literature families and their effect on priority.

Area	Closest sources	Audit consequence
Null pair and double-null GR	Epp; Shyam; Maciel et al.	Timelike/spacelike reconstruction, reciprocal boost scaling and null cross-normalization are established.
Mimetic gravity	Chamseddine–Mukhanov; Shen et al.; Mansoori et al.; Takahashi–Kobayashi	The exact $h = Cg$ map is mimetic; singular composite gravity has known extra-mode and stability hazards.
Cross-flowing radiation	Horvath–Kovacs–Gergely	Two null streams defining an internal time is direct prior art.
Reference matter	Brown–Kuchar; Erlich; Chaurasia et al.	Dynamical clock/ruler fields and emergent composite metrics are established programmes.
Operational null coordinates	Coll–Pozo	Null broadcasts can generate physical coordinates, although proper clocks are inputs there.
Scalar media and superfluids	Celoria et al.; Haber et al.; Samuelsson et al.	The repaired model is a polynomial special case of the known two-superfluid master function $\Psi(p_1^2, p_2^2, p_{12}^2)$. Only the crossed-null state and restricted sum rules remain candidate distinctions.
Null electromagnetism	Kedia et al.; Wylleman–Costa–Natário	Two complex Bateman potentials describe null Maxwell fields, while non-null fields select principal observer families. This is close interpretively but not the same phase-gradient system.
Weyl clocks	Fatibene et al.; Avalos et al.	Clock protocols fixing conformal gauge and second-clock constraints are established.
Causal reconstruction	EPS literature; Braun	Causal order fixes conformal structure; an additional scale/volume datum is required.
Scale generation	Einhorn–Jones; Ji	Dimensional transmutation is established; coupling it to crossed phases may still be distinctive.

Area	Closest sources	Audit consequence
EFT positivity	Adams et al.; de Rham et al.	The healthy sign is expected from analytic local UV completion and positive mediator exchange.

C Conventions and dimensional analysis

The metric signature is $(-, +, +, +)$ and $c = \hbar = 1$ except where stated. The compact phases are dimensionless:

$$[\phi_{\pm}] = 0, \quad [\nabla_a] = 1, \quad [Y_{\pm}] = [C] = 2, \quad [F] = [m^2] = 2, \quad [\kappa] = 0, \quad [H] = 1, \quad [g_H] = 1.$$

The crossed background uses covectors

$$\bar{p}_a = m(1, -1, 0, 0), \quad \bar{q}_a = m(1, +1, 0, 0),$$

so

$$\bar{p}^2 = \bar{q}^2 = 0, \quad \bar{p} \cdot \bar{q} = -2m^2, \quad \bar{C} = m^2.$$

References

- [1] A. Maciel, M. Le Delliou and J. P. Mimoso, “New insights on null and timelike warped symmetric spacetime splittings,” *arXiv:2405.09968* (2024).
- [2] R. J. Epp, “The Symplectic Structure of General Relativity in the Double-Null (2+2) Formalism,” *arXiv:gr-qc/9511060* (1995).
- [3] V. Shyam, “Scale Invariance in Gravity On The Light-Front,” *arXiv:1507.08493* (2015).
- [4] A. H. Chamseddine and V. Mukhanov, “Mimetic Dark Matter,” *JHEP* 11 (2013) 135, *arXiv:1308.5410*.
- [5] L. Shen, Y. Zheng and M. Li, “Two-field mimetic gravity revisited and Hamiltonian analysis,” *JHEP* 12 (2019) 100, *arXiv:1909.01248*.
- [6] S. A. H. Mansoori, A. Talebian, Z. Molaei and H. Firouzjahi, “Multi-field Mimetic Gravity,” *Phys. Rev. D* 105 (2022) 023529, *arXiv:2108.11666*.
- [7] K. Takahashi and T. Kobayashi, “Extended mimetic gravity: Hamiltonian analysis and gradient instabilities,” *JCAP* 11 (2017) 038, *arXiv:1708.02951*.
- [8] S. Hirano, S. Nishi and T. Kobayashi, “Healthy imperfect dark matter from effective theory of mimetic cosmological perturbations,” *JCAP* 07 (2017) 009, *arXiv:1704.06031*.
- [9] Z. Horvath, Z. Kovacs and L. A. Gergely, “Geometrodynamics in a spherically symmetric, static crossflow of null dust,” *Phys. Rev. D* 74 (2006) 084034, *arXiv:gr-qc/0605116*.
- [10] Z. Kovacs, Z. Horvath and L. A. Gergely, “Canonical Analysis of Radiating Atmospheres of Stars in Equilibrium,” *arXiv:gr-qc/0612052* (2006).
- [11] J. D. Brown and K. V. Kuchar, “Dust as a Standard of Space and Time in Canonical Quantum Gravity,” *Phys. Rev. D* 51 (1995) 5600, *arXiv:gr-qc/9409001*.
- [12] B. Coll and J. M. Pozo, “Relativistic Positioning Systems: The Emission Coordinates,” *Class. Quant. Grav.* 23 (2006) 7395, *arXiv:gr-qc/0606044*.
- [13] J. Erlich, “Stochastic Emergent Quantum Gravity,” *Class. Quant. Grav.* 35 (2018) 245005, *arXiv:1807.07083*.

- [14] S. Chaurasia, J. Erlich and Y. Zhou, “Curved Backgrounds in Emergent Gravity,” *Class. Quant. Grav.* 35 (2018) 115008, *arXiv:1710.07262*.
- [15] M. Celoria, D. Comelli and L. Pilo, “Fluids, Superfluids and Supersolids: Dynamics and Cosmology of Self Gravitating Media,” *JCAP* 09 (2017) 036, *arXiv:1704.00322*.
- [16] A. Haber, A. Schmitt and S. Stetina, “Instabilities in relativistic two-component (super)fluids,” *Phys. Rev. D* 93 (2016) 025011, *arXiv:1510.01982*.
- [17] L. Samuelsson, C. S. Lopez-Monsalvo, N. Andersson and G. L. Comer, “Relativistic Two-stream Instability,” *Gen. Rel. Grav.* 42 (2010) 413, *arXiv:0906.4002*.
- [18] H. Kedia, I. Bialynicki-Birula, D. Peralta-Salas and W. T. M. Irvine, “Tying knots in light fields,” *Phys. Rev. Lett.* 111 (2013) 150404, *arXiv:1302.0342*.
- [19] L. Wylleman, L. F. O. Costa and J. Natário, “Poynting vector, super-Poynting vector, and principal observers in electromagnetism and general relativity,” *Class. Quant. Grav.* 38 (2021) 165009, *arXiv:2007.15384*.
- [20] L. Fatibene, S. Garruto and M. Polistina, “Breaking the Conformal Gauge by Fixing Time Protocols,” *Int. J. Geom. Meth. Mod. Phys.* 12 (2015) 1550044, *arXiv:1410.1284*.
- [21] R. Avalos, F. Dahia and C. Romero, “A note on the problem of proper time in Weyl space-time,” *Found. Phys.* 48 (2018) 253, *arXiv:1611.10198*.
- [22] A. Connes and C. Rovelli, “Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories,” *Class. Quant. Grav.* 11 (1994) 2899, *arXiv:gr-qc/9406019*.
- [23] M. Braun, “Spacetime reconstruction by order and number,” *arXiv:2507.01907* (2025).
- [24] N. Linnemann and J. Read, “Constructive Axiomatics in Spacetime Physics Part I: Walkthrough to the Ehlers–Pirani–Schild Axiomatisation,” *arXiv:2112.14063* (2021).
- [25] A. Adams, N. Arkani-Hamed, S. Dubovsky, A. Nicolis and R. Rattazzi, “Causality, Analyticity and an IR Obstruction to UV Completion,” *JHEP* 10 (2006) 014, *arXiv:hep-th/0602178*.
- [26] C. de Rham, S. Melville, A. J. Tolley and S.-Y. Zhou, “Positivity Bounds for Scalar Theories,” *Phys. Rev. D* 96 (2017) 081702, *arXiv:1702.06134*.
- [27] M. B. Einhorn and D. R. T. Jones, “Naturalness and Dimensional Transmutation in Classically Scale-Invariant Gravity,” *JHEP* 03 (2015) 047, *arXiv:1410.8513*.
- [28] X. Ji, “Scale symmetry breaking, quantum anomalous energy and proton mass decomposition,” *Nucl. Phys. B* 960 (2020) 115181, *arXiv:2105.03974*.
- [29] J. E. Juknevič, “Pure-gluon hidden valleys through the Higgs portal,” *JHEP* 08 (2010) 121, *arXiv:0911.5616*.
- [30] D. M. Ghilencea, “Stueckelberg breaking of Weyl conformal geometry and applications to gravity,” *Phys. Rev. D* 101 (2020) 045010, *arXiv:1904.06596*.